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Axially Perpendicular Offset Raman Scheme for Reproducible Measurement of Housed Samples in a Noncircular Container under Variation of Container Orientation

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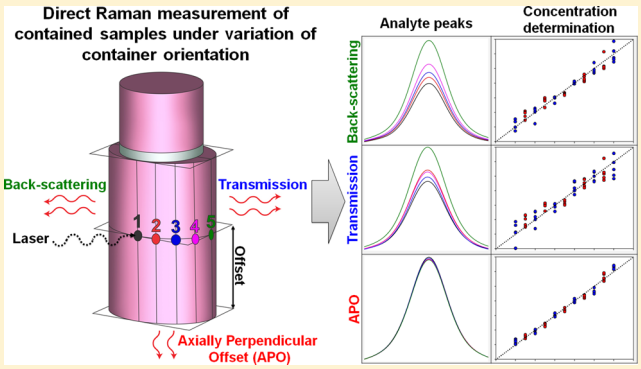
ABSTRACT: An axially perpendicular offset (APO) scheme that is able to directly acquire reproducible Raman spectra of samples contained in an oval container under variation of container orientation has been demonstrated. This scheme utilized an axially perpendicular geometry between the laser illumination and the Raman photon detection, namely, irradiation through a sidewall of the container and gathering of the Raman photon just beneath the container. In the case of either backscattering or transmission measurements, Raman sampling volumes for an internal sample vary when the orientation of an oval container changes; therefore, the Raman intensities of acquired spectra are inconsistent. The generated Raman photons traverse the same bottom of the container in the APO scheme; the Raman sampling volumes can be relatively more consistent under the same situation. For evaluation, the backscattering, transmission, and APO schemes were simultaneously employed to measure alcohol gel samples contained in an oval polypropylene container at five different orientations and then the accuracies of the determination of the alcohol concentrations were compared. The APO scheme provided the most reproducible spectra, yielding the best accuracy when the axial offset distance was 10 mm. Monte Carlo simulations were performed to study the characteristics of photon propagation in the APO scheme and to explain the origin of the optimal offset distance that was observed. In addition, the utility of the APO scheme was further demonstrated by analyzing samples in a circular glass container.



Raman spectroscopy is a useful analytical method for the fast and nondestructive analysis of a diverse range of samples.[1−6] It is especially useful when rapid, nondestructive quality assurance of the samples is necessary. Recently, a popular application for Raman spectroscopy is the direct analysis of samples housed in containers without the need to manually transfer samples; this application is convenient and avoids sample contamination.[7−10] Matousek et al. and other research groups have reported spatially offset Raman scattering (SORS)-based schemes[11−14] that are able to collect the Raman spectra of samples (beverages, medicines, and cosmetics) in several different containers made of glass or polymers with little to no overlapping between the sample and container peaks. Such measurements enabled simple and accurate identification of the housed samples.[3,15−21] In addition, stand-off SORS schemes that are able to collect the spectra of samples in containers at a remote distance have also been demonstrated. For example, the spectral acquisition of $NaClO_3$ and isopropyl alcohol, placed inside a 1.5 mm-thick low-density polyethylene (LDPE) bottle at a distance of 12 m, is possible.[22] Alternatively, time-resolved Raman spectroscopy (TRRS), equipped with a picosecond pulsed laser, was useful to separate Raman features of internal samples from those of polytetrafluoroethylene (PTFE), polyoxymethylene (POM), and polyethylene (PE) containers.[23]

The above-listed reports describe the successful identification of materials housed in diverse containers. At the same time, the nonsampling quantitative analysis of contained samples has become more demanding since it is highly versatile for nondestructive analysis of many practical field samples. For accurate quantitative analysis, the acquisition of reproducible and quantitatively representative sample spectra is essential. Unfortunately, spectral features of housed samples susceptibly vary under several situations such as variation of the laser illumination spot on a container and a change in the container orientation. Previously, our research group demonstrated the determination of the concentrations of povidon and hydrogen peroxide in circular PE bottles using the wide area illumination (WAI) scheme, which provided a wide area (28.3 mm^2) of laser irradiation with a long focal length (25 cm). The accuracy in the determination of the hydrogen peroxide concentration was not seriously degraded within ±5 mm positional variations of the container. Since the samples measured in these studies were

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in circular bottles, the variation of bottle orientation did not seriously deteriorate the reproducibility of the measurements.[9] However, when the shape of a container is not circular (such as oval-shaped containers), changes in the bottle orientation can generate quantitatively inaccurate spectra of a contained sample. This is due to the variation of the sampled volume at spectral acquisition and the difference in the radius curvature of the wall, which can alter the propagation of photons. Therefore, an analytical strategy that can collect reproducible Raman spectra of a contained sample, regardless of changes in the container orientation, is in great demand.

We have demonstrated the so-called axially perpendicular offset (APO) scheme for the direct acquisition of reproducible Raman spectra of samples contained in an oval container under variation of container orientation. The APO scheme is based on an axially perpendicular geometry between the laser illumination and Raman photon detection as depicted in Figure 1a.

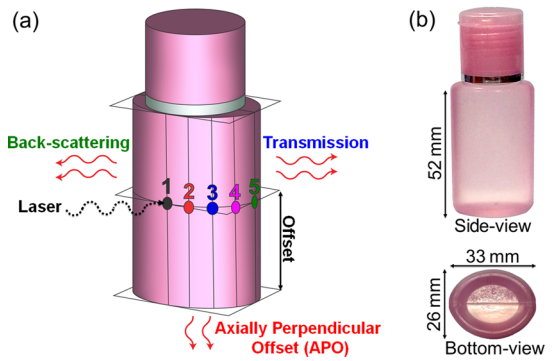

Figure 1. (a) Five different laser-illuminated spots on the container for Raman measurements and the offset distance in the APO measurement are depicted. The directions of laser illumination and photon detection in the backscattering, transmission, and APO schemes are also described. (b) Side- and bottom-view images of the oval PP container retaining a 66.0% alcohol gel.

This setup corresponds to laser irradiation through the sidewall of an oval container, while photon detection occurs beneath the bottom of the container. In this scheme, the sampled volumes for spectral acquisition can have relatively little variation. Additionally, the internally generated Raman photons cross the same bottom barrier for detection, so the acquired spectra of an internal sample are expected to be more reproducible. To evaluate the potential of the APO scheme, the Raman spectra of alcohol (ethanol) gels housed in an oval polypropylene (PP) container were collected at five different container orientations, as shown in Figure 1a, and an optimal axial offset distance yielding the most reproducible spectral features was searched for between 4 and 20 mm. With the acquired spectra, the concentrations of alcohol were determined using partial least-squares (PLS),[24−26] and the influence of the change in the container orientation on the measurement accuracy was investigated. Furthermore, the propagation of Raman photons in the APO scheme was studied using a Monte Carlo method, and the calculated reproducibility, according to the axial offset distance, was compared with the experimental results. For comparison, the backscattering and transmission Raman spectra collected under the same conditions were also used to determine the alcohol concentrations and the results were compared with those achieved using the APO scheme. Finally, the analytical utility of the APO scheme was further demonstrated by analyzing samples in a circular glass container: the most common type of sample container.

■ EXPERIMENTAL SECTION

Sample Preparation. Figure 1b shows the side- and bottom-view pictures of the oval container used in this study. The longest and shortest widths of the container body were 33 and 26 mm, respectively, and the container had a height of 52 mm. The body was made of translucent PP with a pink color. Eleven alcohol gel samples (concentration range: 56.0−76.0 wt % with an increment of 2 wt %) were prepared by mixing the appropriate amounts of pure ethanol and gel base. Eleven different containers with the same size were used to separately house the prepared alcohol gel samples of different concentrations. Upon preparation, the sample-housed containers were tightly sealed to prevent possible ethanol evaporation.

Acquisition of Raman Spectra of Alcohol Gels Housed in an Oval Container. For the backscattering measurement, a spectrometer (RamanRxn1 unit, Kaiser Optical Systems, Ann, Arbor, MI, USA) equipped with a PhAT probe that provided a large illumination area (28.3 mm^2; diameter: 6 mm) and a wide photon-acceptance area (506.7 mm^2; diameter: 25.4 mm) was used. It had 25.4 cm working distance with ±10.0 cm depth of field. In the case of the transmission and APO measurements, a separate laser (Invictus, Kaiser) was used for excitation and the Raman spectra were collected using the same probe and spectrometer. The wavelength of the laser was 785 nm for all three schemes. The probe was located at 24.8 cm away from a PP container for Raman photon detection. The APO scheme was similarly based on traditional 90° excitation geometry, while it differed from a conventional scheme since the detection of Raman photon was axially accomplished beneath a container at optimized offset distance using a detector providing a wide photon-acceptance area.

In each case, the Raman spectra of the contained alcohol gel samples were directly collected by rotating the container to illuminate the laser at five different locations on the side, as described in Figure 1a. The laser exposure times were 5, 20, and 50 s for the backscattering, transmission, and APO measurements, respectively. The resolution of the collected spectra was 4 cm^{-1}. Baseline correction of the raw spectra and PLS were performed using MATLAB version 7.0 (MathWorks Inc., MA, USA). A Monte Carlo simulation was performed using fully licensed Monte Carlo (Version 2014).[27,28]

■ RESULTS AND DISCUSSION

Figure 2 shows the Raman spectra of an empty oval PP container (black), a 66.0% alcohol gel in a quartz cuvette (red), and the same sample in the container (blue). The spectra were collected using the backscattering scheme. The spectra are arbitrarily offset to enable an easier comparison. As readily apparent, the spectrum of the alcohol gel housed in the container includes the combined spectral features of the empty container and the alcohol gel. There are five major ethanol peaks at 1452, 1274, 1045, 878, and 434 cm^{-1}; the 878 cm^{-1} band (C−C stretching)[29] is the most intense and has minor overlapping with the PP peaks. Since the spectral acquisition will be accomplished directly through the container, the intensity ratio between an alcohol peak and a container peak in the acquired spectra must be consistent in any/all cases to ensure the accurate determination of the alcohol concentrations.

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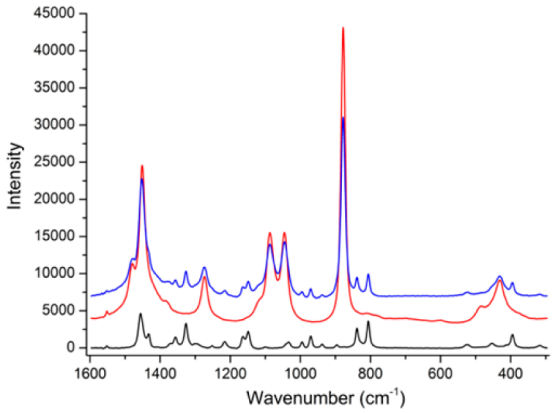

Figure 2. Backscattering Raman spectra acquired from an empty oval PP container (black), a 66.0% alcohol gel in a quartz cuvette (red), and the same sample in the oval PP container (blue).

Using both the backscattering and transmission schemes, Raman spectra of a 66.0% alcohol gel in the container were collected by changing the locations of laser illumination, as described in Figure 1a. The radius curvatures at spots #1 and #5 are the smallest and largest, respectively. Figure 3a shows the acquired backscattering Raman spectra of the sample. The colors of each spectrum distinguish the illuminated spots for spectral collection (#1: black; #2: red; #3: blue; #4: pink; #5: green). The spectral variations of the alcohol band at 878 cm^{-1} and the PP band at 806 cm^{-1} are separately magnified in the insets. It was observed that the baselines of the spectra varied between measurements. Figure 3b shows the corresponding baseline-corrected spectra. The baselines were linearly corrected at 925, 750, 700, and 350 cm^{-1}. As also shown in the inset, the intensities of the alcohol band vary somewhat between the measurements. The Raman peaks of the alcohol gel are most intense at spot-#5 and subsequently decrease as the illumination spot moves from #5 to #1. In the case of illumination at spot-#5, the distance between the illuminated face and the opposite face is the longest (33 mm); therefore, a relatively large amount of alcohol gel is exposed within the Raman sampling volume, which yields the higher peak intensity. In contrast, the intensities are smallest when spot #1 is illuminated; this is due to the decreased sampled volume (face-to-face distance: 26 mm).

To analyze the spectral variations quantitatively, relative standard deviations (RSDs) of the intensities of the raw spectra at each wavenumber (black) were calculated, as shown in Figure 4a. The large offset in the RSD (approximately 6.3%) originates from the baseline variation of the spectra. RSDs are substantially higher at the wavenumbers where the major alcohol bands are located. Furthermore, the intensity ratio ($I_{878\ cm^{-1}}/I_{806\ cm^{-1}}$) between the 878 cm^{-1} (ethanol) and 806 cm^{-1} (PP container) bands was calculated using the baseline-corrected spectra, as shown in Figure 4b. Steady $I_{878\ cm^{-1}}/I_{806\ cm^{-1}}$ values among the measurements imply that the intensities of both peaks change in proportion to one another. When this occurs, the quantitative representation of the alcohol concentration in the spectra is acceptable. The $I_{878\ cm^{-1}}/I_{806\ cm^{-1}}$ values increase when the laser-irradiated spots vary from #1 to #5, and the RSD of these $I_{878\ cm^{-1}}/I_{806\ cm^{-1}}$ values was 11.0%; the accuracy for the determination of the alcohol concentration

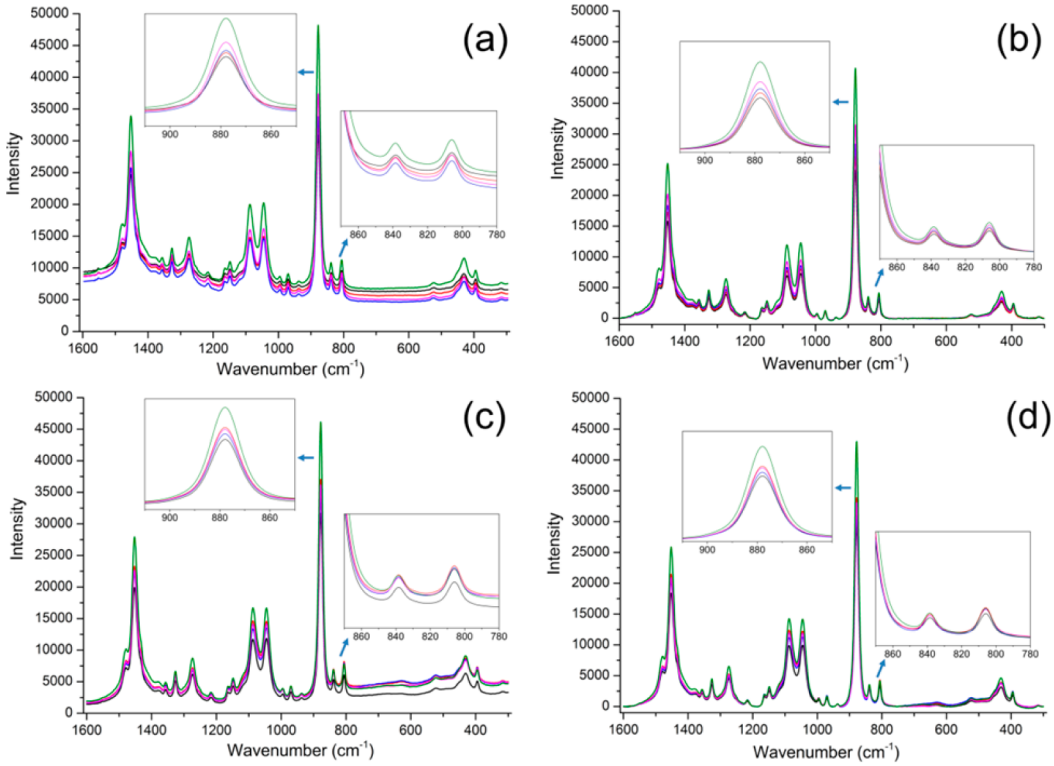

Figure 3. (a) Backscattering and (c) transmission Raman spectra of a 66.0% alcohol gel in the oval PP container acquired with changes in the container orientation. The colors of the spectra distinguish the illuminated spots used for spectral collection (#1: black; #2: red; #3: blue; #4: pink; #5: green; refer to Figure 1). Baseline-corrected backscattering (b) and transmission (d) are also displayed. In all cases, the alcohol band at 878 cm^{-1} and the PP band at 806 cm^{-1} are separately highlighted in the insets.

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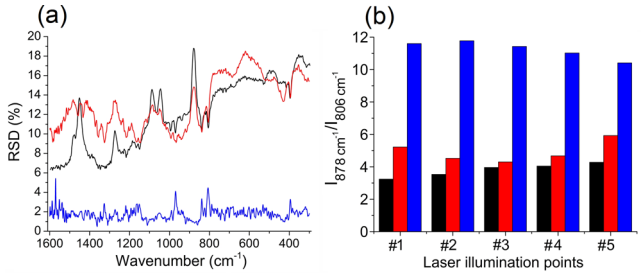

Figure 4. (a) The RSDs of the intensities at each wavenumber over the spectral range calculated from the backscattering (black), transmission (red), and APO Raman spectra (blue) of a 66.0% alcohol gel acquired with changes in the container orientation. (b) The intensity ratios ($I_{878\ cm^{-1}}/I_{806\ cm^{-1}}$) between the 878 cm^{-1} (ethanol) and 806 cm^{-1} (PP container) bands calculated from the backscattering (black), transmission (red), and APO Raman spectra (blue) are also shown.

appears to degrade when the backscattering measurement is employed.

Figure 3c,d shows the transmission and corresponding baseline-corrected Raman spectra of the same sample collected with changes in the container orientation, respectively. The 878 and 806 cm^{-1} bands are magnified in the insets. As shown in Figure 4a, the RSDs of the intensities at each wavenumber fluctuate over the range and a large offset is again observed. The calculated $I_{878\ cm^{-1}}/I_{806\ cm^{-1}}$ values (Figure 4b) indicate inconsistent transmission measurements with an RSD of 13.2%. These results are unsatisfactory for accurate concentration determination.

Optimal Offset Distance in the APO Measurement. The variation of axial offset distance in the APO scheme can directly influence the spectral features of the acquired spectra. To investigate the possible spectral variation, the offset distance was increased from 4 to 20 mm at an interval of 2 mm and the Raman spectra of the same sample were collected with changes in the container orientation at each offset distance. Figure 5a−d shows the baseline-corrected APO spectra that were collected at the offset distances of 6, 10, 14, and 18 mm, respectively. The 878 and 806 cm^{-1} bands are again magnified in the insets. As shown, the intensities of both the alcohol and PP bands decreased as the offset distance increased. This decrease in the Raman signal is expected, due to the decreased chance that Raman photons will reach the detector as the offset distance increases.

One noticeable observation is that the intensities of some bands, such as the alcohol gel peak at 878 cm^{-1} with an offset distance of 10 mm, are the most reproducible. To examine the reproducibility of the spectral features at all of the tested offset distances, the average intensities of the 878 cm^{-1} band (circles) and corresponding RSDs (squares) are compared, as shown in Figure 6a. The average intensity decreases with a longer offset distance as expected, while the RSD is suddenly minimized at the offset distance of 10 mm (with an RSD of 1.0%). Figure 4a also shows the RSDs of the intensities over the spectral range (black line) from the APO measurement at the offset distance of 10 mm. The RSDs are substantially smaller compared to those obtained using the backscattering and transmission schemes. $I_{878\ cm^{-1}}/I_{806\ cm^{-1}}$ values obtained in the APO measurement are also shown in Figure 4b. The magnitudes of the $I_{878\ cm^{-1}}/I_{806\ cm^{-1}}$ values are substantially greater compared to the other two measurements. A higher $I_{878\ cm^{-1}}/I_{806\ cm^{-1}}$ value is preferable because information about the alcohol gel becomes stronger as the intensity of the container peaks is minimized. The RSD of the $I_{878\ cm^{-1}}/I_{806\ cm^{-1}}$ values decreased to 4.8%. The APO measurement produces spectra with reproducible

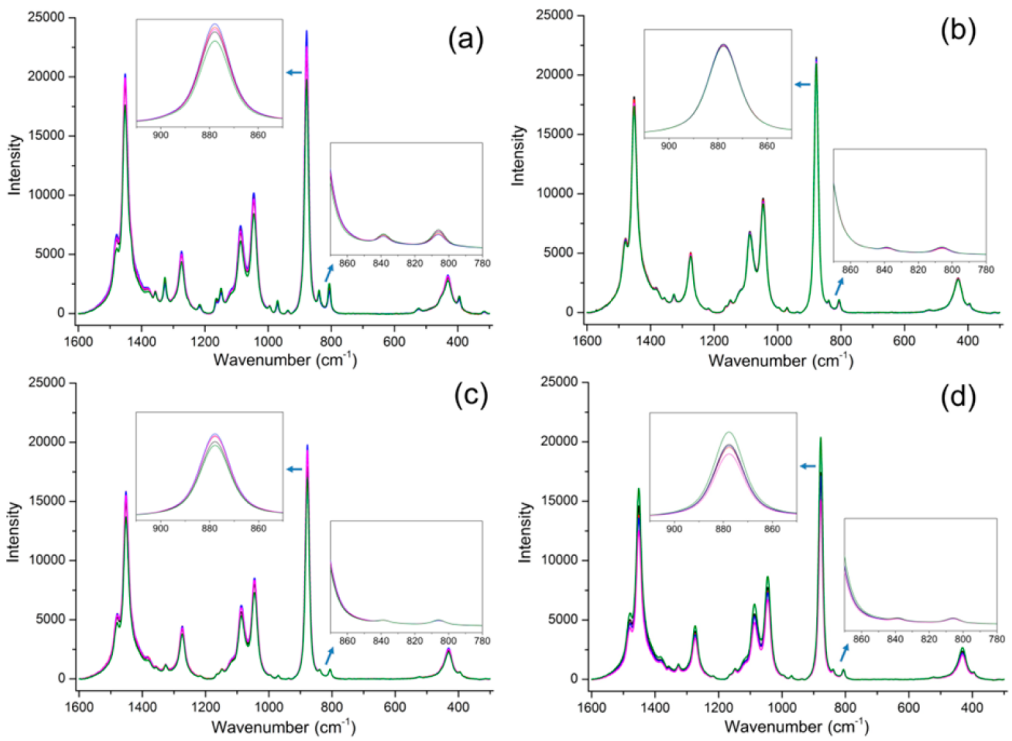

Figure 5. Baseline-corrected APO spectra of a 66.0% alcohol gel acquired at five different container orientations when the offset distance is (a) 6, (b) 10, (c) 14, and (d) 18 mm. The 878 and 806 cm^{-1} bands are magnified in the insets.

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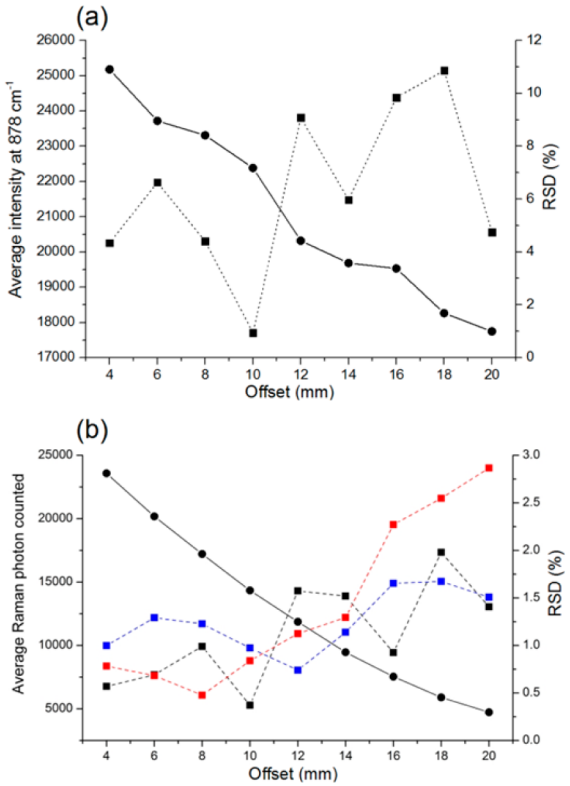

Figure 6. (a) Average intensities (circles) of the 878 cm^{-1} band and the corresponding RSDs (squares) obtained from actual APO Raman measurements and (b) Monte Carlo simulations. The optimal offset distances obtained by the simulation when the size of the container decreases (red squares) and increases (blue squares) by 10% are also shown.

quantitative information under changes in the container orientation. Therefore, the offset distance was fixed at 10 mm for further APO measurements.

Determination of the Alcohol Concentrations of Samples Housed in Oval Containers. To measure alcohol concentrations, a PLS model was built using six calibration samples, and the constructed model was used to predict the concentrations of five validation samples. Each sample had five spectra acquired at five different container orientations, as described in Figure 1. The concentration scatter plots of the alcohol concentrations determined by backscattering, transmission, and APO measurements are shown in Figure 7a−c, respectively. Blue and red circles correspond to the calibration and validation data, respectively. In all cases, five PLS factors were used for the modeling. As shown, the calibration and prediction data points are much closer to the diagonal line in the APO measurement. The standard error of calibrations (SECs) and standard error of predictions (SEPs) were 1.68% and 1.47%, 2.16% and 1.89%, and 0.86% and 0.75%, for the backscattering, transmission, and APO measurements, respectively.

To analyze the dependency of the prediction accuracy on the container orientation, a PLS model was again developed. This used all of the sample spectra collected at spot #1 for the modeling, and the model was used to predict the concentrations of the same samples measured at the other spots (#2, #3, #4, and #5). Table 1 shows the resulting SEPs, prediction bias, and SEPs after adjustment of the prediction bias (bias-adjusted SEP). In the backscattering and transmission measurements, both SEPs and prediction biases continuously increase as the spectra are collected at spots farther from spot #1. They were greatest when the spectra obtained at spot #5 were used to predict the alcohol concentrations. The occurrence of prediction bias indicates that there is a systematic spectral difference between the spectra acquired at the five different container orientations, so the bias-adjusted SEPs mostly decreased. Nonetheless, the bias-adjusted SEPs remain considerably high at the #4- and #5-spot illuminations in the backscattering measurement and the #5-spot illumination in the transmission measurement. This indicates that, in addition to the systematic spectral difference, there are more spectral discrepancies in these cases. In contrast, the variations of SEPs and prediction biases are smaller in the APO measurement. When the prediction biases are adjusted, the errors decrease substantially. This informs us that minor systematic spectral differences remain among the spectra, and these are unable to be eliminated even with the use of the APO scheme.

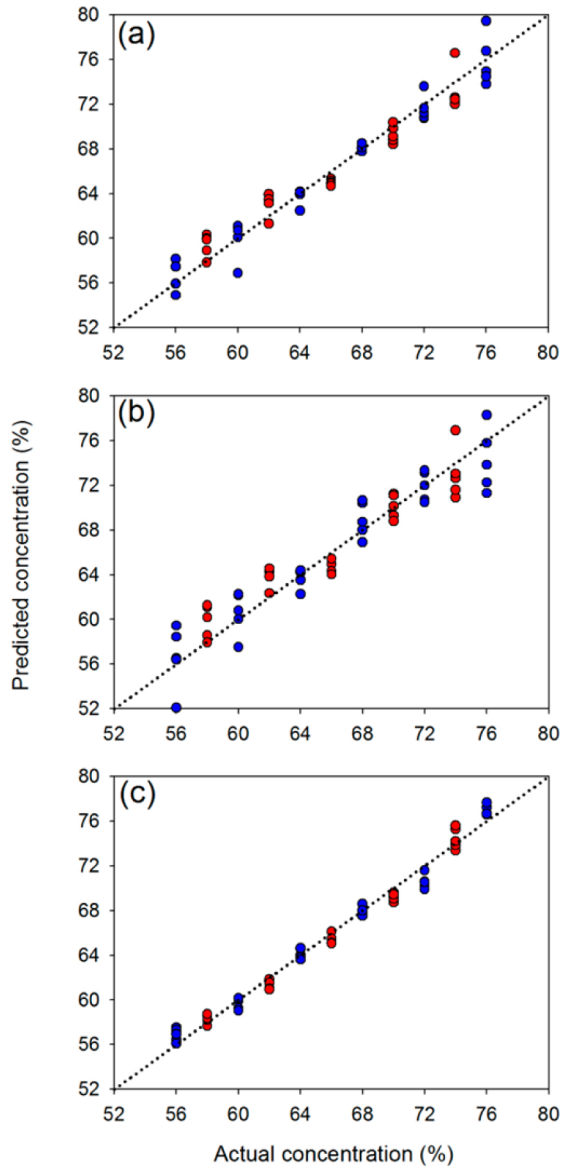

Figure 7. Concentration scatter plots from the determination of the alcohol concentrations when the (a) backscattering, (b) transmission, and (c) APO measurements are employed. Blue and red circles correspond to the calibration and validation data, respectively.

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Table 1. SEPs from the Prediction of Alcohol Concentrations Using the Spectra Acquired by Laser Illuminations at Spots #2, #3, #4, and #5[a]

Raman measurement	laser illuminated spot	SEP, %	bias, %	bias-adjusted SEP, %
backscattering	#2	0.50	−0.14	0.48
	#3	0.76	−0.19	0.73
	#4	1.94	−0.76	1.79
	#5	5.66	−3.29	4.61
transmission	#2	3.18	−2.30	0.45
	#3	3.86	−2.76	0.59
	#4	4.10	−2.99	0.73
	#5	4.36	−3.51	1.42
APO	#2	0.94	0.33	0.23
	#3	0.79	0.58	0.27
	#4	0.80	0.58	0.10
	#5	0.65	0.22	0.15

[a]A PLS model was developed using the spectra collected with the spot #1-illumination. Prediction bias and SEPs after adjustment of the prediction bias are also shown.

Nonetheless, the APO scheme provides more reproducible spectra, thereby increasing the accuracy for the determination of the alcohol concentrations.

Simulation of Photon Propagation in the APO Scheme Using a Monte Carlo Method. To probe the origin of the experimentally found optimal offset distance, the characteristics of Raman photon propagation in the APO scheme were investigated using a Monte Carlo method. For this purpose, three separate boundaries for the sidewall of the PP container, housed alcohol gel, and bottom of the container were set. The internal distance between opposite walls (or sample thickness in Raman measurement) varied between 26.00, 27.75, 29.50, 31.25, and 33.00 mm, which correspond to the laser illumination at spots #1, #2, #3, #4, and #5, respectively. The diameter of laser illumination was 1 mm. It was assumed that individual photons propagated independently through the sample and the two PP barriers in a random walk-like fashion in three-dimensional space. The size of the detecting window was 2.54 cm in diameter, which is the same size as the probe used for actual measurements. Only the Raman photons reaching the detector window were counted.

For the simulation, several important parameters such as mean free path and absorption coefficient of the photon-crossing medium need to properly set. Mean free path (λ) was determined using the following equation as previously reported[30,31]

$$\lambda = \frac{RT}{\sqrt{2}\,\pi d^2 N_A P}$$

where R, T, d, N_A, and P corresponds to universal gas constant (8.3145 J/mol·K), temperature (K), molecular diameter (m), Avogadro's number (6.023×10^{23}), and pressure (Pa), respectively. For the calculation of molecular diameter, the mass of a molecule was initially calculated by dividing its molar mass by Avogadro's number and volume of the molecule was

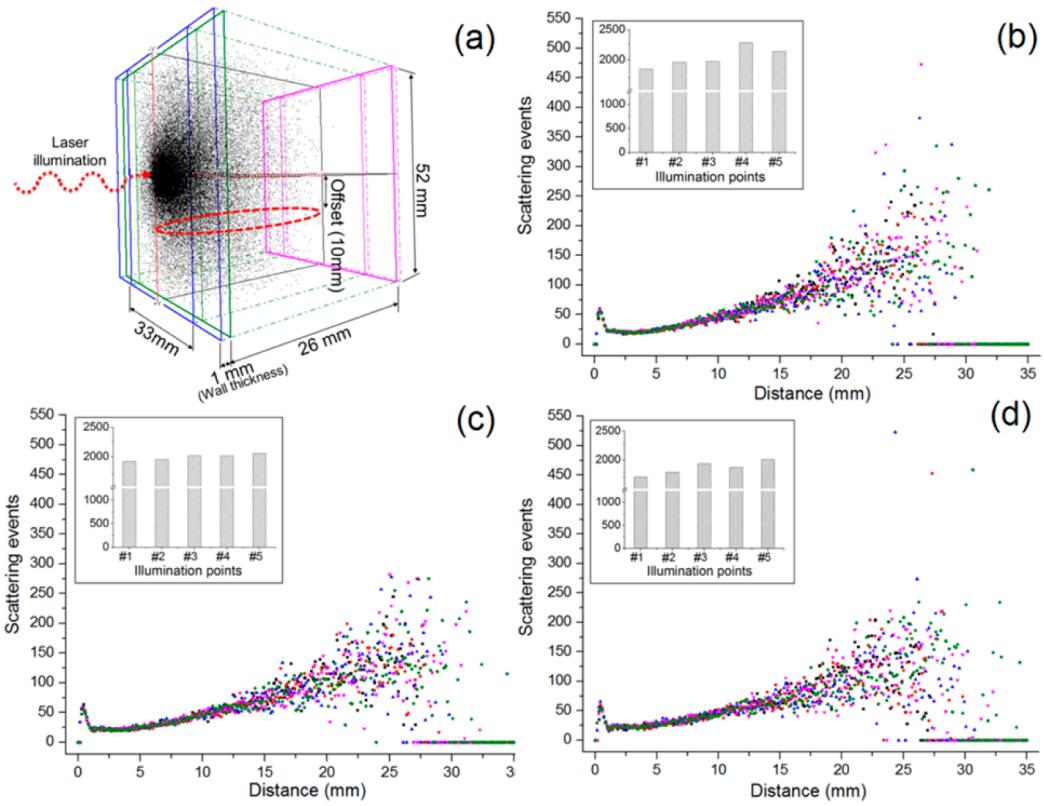

Figure 8. (a) Distribution of Raman photons (black dots) in a space when the laser is irradiated on the wall of the PP container housing a 66.0% alcohol gel. The wavy arrow indicates the point of irradiation, and the dashed oval depicts a photon-receiving area (2.54 cm in diameter) of a detector at an offset distance of 10 mm. The variation of the Raman scattering events arriving over a horizontal plane at (b) 6, (c) 10, and (d) 14 mm offset distances when the laser illuminates five different spots on the oval container. The data point colors indicate the illuminated spots (#1: black; #2: red; #3: blue; #4: pink; #5: green). The insets show the integrated Raman scattering events at each measurement.

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then calculated using the corresponding density. Next, diameter of the molecule was obtained by assuming that the shape of molecule was spherical. The mean free paths for PP, glycerol (gel matrix), and ethanol calculated by the equation were 1.57, 24.2, and 28.1 nm, respectively. The absorption coefficient (ξ) of ethanol at 785 nm was 0.010500 cm^{-1},[32] while those of PP and glycerol were set to zero due to no absorption of 785 nm radiation. The refractive indices of PP, glycerol, and ethanol were set to 1.485, 1.470, and 1.350, respectively. The tested sample housed in the PP container was composed of 66.0% ethanol and 34.0% glycerol.

Figure 8a shows the distribution of Raman photons (black dots) in a space when laser irradiates on the wall of a PP container housing a 66.0% alcohol gel. The wavy arrow indicates the point of laser irradiation. Raman photons are isotropically spread in the space, and the photon population density decreases as the distance becomes further away from the point of laser illumination. The dashed circle (looks like an oval) depicts a photon-receiving area (2.54 cm in diameter) of a detector at an offset distance of 10 mm. As expected, the larger number of photons arrives at the horizontal detector plane when an offset distance becomes shorter.

Figure 8b shows the variation of the Raman scattering events occurring over a horizontal plane at a 6 mm offset. The distance in the x-axis implies the horizontal distance from the point of laser irradiation. The colors designate the five different illumination spots on the oval container for measurements. A higher Raman scattering event at a given distance implies the arrival of more Raman photons. The Raman scattering events gradually increase up to the distance of 14 mm, and the chances for events are similar for all five cases. Beyond 14 mm, the number of events starts to decrease and greatly fluctuate between the measurements due to the increased travel distance from the excitation point (equivalently, increased uncertainty in photon propagation). More consistent Raman scattering events among the five measurements directly imply the acquisition of more reproducible Raman spectra as the container orientation is changed. The integral of the Raman scattering events (photons) at the detector area will be the final Raman intensity of the corresponding measurement. The results of integrations for the measurements are displayed in the inset. The integrated intensities somewhat vary among the measurements as shown.

Figure 8c,d shows the Raman scattering events that occurred in the measurements at the offset distances of 10 and 14 mm, respectively. In comparison with the results obtained at the 6 mm offset, the overall number of Raman scattering events decreased due to the detection of photons at the farther locations, while the maximum occurrence is observed around a distance of 12 mm (in the case of 10 mm offset). A noticeable observation in the 10 mm offset measurement is that the degree of event dispersion in the 14−26 mm range is relatively smaller compared to those of the other measurements. This indicates that the overall amount of Raman photons perceived by a detector at each orientation is relatively less variable at the 10 mm offset as also observable in the corresponding inset showing the integrated Raman scattering events at each measurement; this agrees with the experimental results that obtained the most reproducible spectra at this condition. To examine the spectral reproducibilities for all of the tested offset distances, the Raman intensities at the five different orientations for each offset distance were averaged and the corresponding RSDs were calculated using the simulation, as shown in Figure 6b. The variations of the average intensity (circles) and RSD (black squares), according to the offset distance, are similar to the experimental results (shown in Figure 6a), and the simulation also draws the lowest RSD at the 10 mm offset distance.

It was expected that the optimal axial offset distance could vary with the size of the oval container. Using the same simulation, we studied how the optimal offset distance varied according to changes in the oval container size (±10% of the original size that was studied). Figure 6b also shows the calculated RSDs at all of the tested offset distances as the size of the container decreases (red squares) or increases (blue squares) by 10%. When the container size decreases by 10%, Raman intensities are most reproducible at an offset distance of 8 mm. This result is reasonable because a shorter offset distance is likely to be required to collect more reproducible Raman photons as the container becomes smaller. Meanwhile, the optimal offset distance increases to 12 mm in the case of the container that is 10% larger. Although actual measurements using real oval containers with different sizes were not accomplished, the simulation results appeared to be reasonable.

It is also interesting to examine the variation of optimal offset distance when absorption coefficient of a housed sample changes. For the investigation, the absorption coefficient of a sample increased from 0.01 to 0.05 to 0.10 cm^{-1}, and the simulation was performed for each case while keeping the other parameters unchanged. The found optimal offset distances by the simulation were 10, 8, and 6 mm when the absorption coefficients were 0.01, 0.05, and 0.10 cm^{-1}, respectively. The optimal offset distance clearly decreases with the increase of absorption coefficient. When the absorption coefficient of a housed sample increases, the distribution of generated Raman photons as shown in Figure 8a becomes smaller due to the attenuation by some degree of absorption, while overall distribution pattern remains the same. Therefore, the shorter optimal offset distances are reasonable when the absorption of photons by a housed sample increases.

When the APO scheme is employed for actual measurement in a field, the variation of container thickness by container-to-container variability that would possibly deteriorate an optimal measurement needs to be considered. When a container wall becomes thicker, the diffusion of laser radiation in the wall becomes greater. This makes the distribution of Raman photons larger, and so, an optimal offset distance would be longer. For the examination, the simulation was performed by changing the wall thickness by ±10% (equivalent to the variation from 0.9 to 1.1 mm) with intervals of 2%, while keeping the other parameters unchanged. When the thickness variation is +2%, the found optimal distances are unchanged (equally 10 mm). When the thickness increases by 4%, the optimal offset distance changes to 12 mm and then remains steady up to a 10% change. On the other hand, the optimal offset distance (10 mm) remains unchanged with the decrease of thickness up to 6% and changes to 12 mm when the thickness varies by −8% as well as −10%. As found by the simulation, the thickness variation starts to make a given APO scheme no longer optimal and eventually degrades the accuracy of concentration determination. Such plastic containers are thoroughly mass-produced, and the variation of container thickness is very small and usually below ±2%; therefore, no significant degradation of accuracy due to the container-to-container variability is expected practically. Nonetheless, it is worthwhile to note that a larger variation of thickness will deteriorate measurement accuracy.

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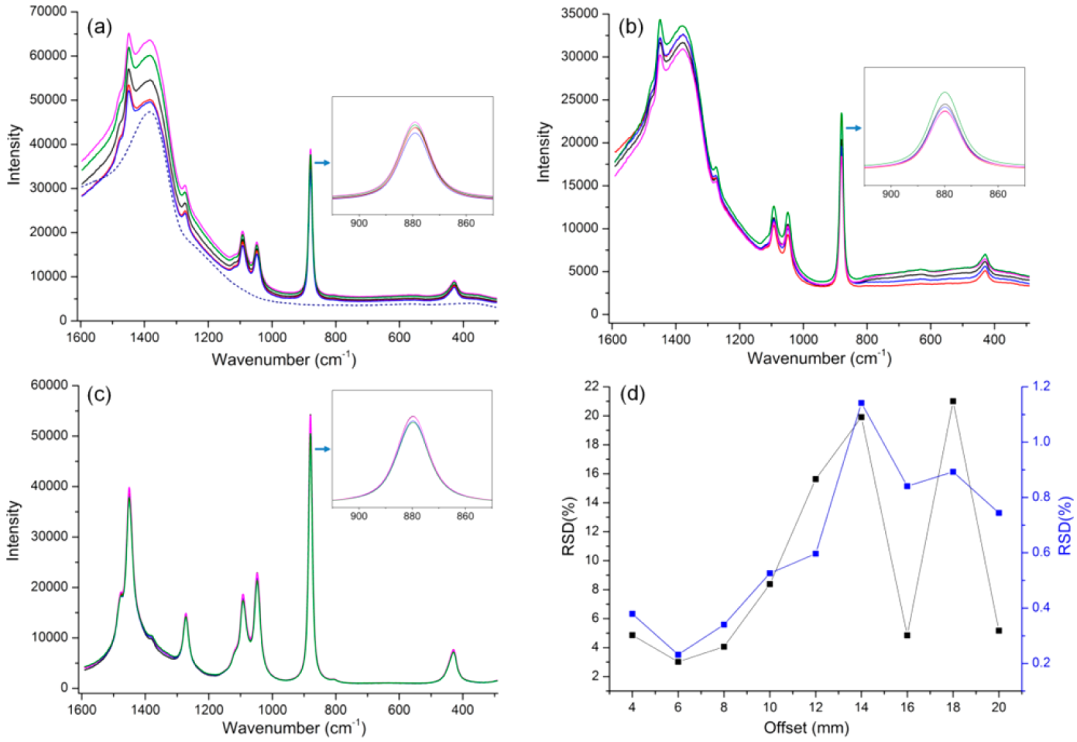

Figure 9. (a) Backscattering, (b) transmission, and (c) APO Raman spectra of a 66.0% alcohol gel in a round glass container acquired with changes in the container orientation. The colors of the spectra distinguish the five different illuminated spots used for spectral collection. The alcohol band at 878 cm^{-1} is highlighted in the insets. (d) The RSDs of the intensities at 878 cm^{-1}, obtained using a 6 mm offset, for the APO measurement (black) and Monte Carlo simulation (blue) are also shown.

APO Scheme for Analysis of a Sample in a Round Glass Container. The analytical usage of the APO scheme was further evaluated for the measurement of a 66.0% alcohol gel in a round glass container (26 mm in diameter, 2 mm in wall thickness). The laser was illuminated at five different spots on the glass container, using the same method described in Figure 1. Because a circular container was used, the sampled volumes at each measurement were nearly constant. For the purpose of comparison, the backscattering and transmission spectra of the same sample were collected in the exact same way. Figure 9a shows the backscattering Raman spectra measured by the illuminating laser at five different spots on the sample contained in the round glass container. The Raman peaks of ethanol in the 1000−300 cm^{-1} range are clearly observable, while those in the 1600−1000 cm^{-1} range are relatively indistinct due to the fact that they overlap with the glass Raman background. The Raman spectrum acquired from the same empty glass container is also shown (dashed spectrum).The magnitudes of the glass background for these measurements vary somewhat due to the variation of the illuminated spots on the glass wall. To examine the variation of the alcohol band intensity, the strongest band (878 cm^{-1}) that is not overlapping with the glass background is highlighted and shown in the inset. The Raman intensities still vary with an RSD of 3.6% when the alcohol gel sample is measured in the circular glass container. Before calculating the RSD, the baselines of the band were corrected. In the case of the transmission measurement (Figure 9 b), the overall Raman intensities decreased due to the loss of Raman photons as they traveled longer distances to the detector. The baselines also vary between the spectra, and the RSD of the band intensity at 878 cm^{-1} is 10.1% after baseline correction.

By changing the offset distance from 4 to 20 mm (at intervals of 2 mm), the APO spectra of the sample were acquired with laser illuminations on the same five spots. Then, the RSDs of the 878 cm^{-1} band at each offset distance (black squares) were calculated, as shown in Figure 9d. The band intensity is most reproducible at the 6 mm offset (RSD: 3.0%). Figure 9c shows the acquired APO spectra at the offset distance of 6 mm. The most evident observation is that the glass background disappears and only the spectral features corresponding to ethanol are clearly visible over the spectral range. The capability of the APO scheme to minimize or eliminate the glass background serves as an additional advantage since the accuracy of the measurement can be improved in the absence of this interfering background Raman signal. In addition, the baselines of the spectra are stable; this represents another advantage for reliable quantitative analysis. Figure 9d also shows the simulation results for the RSDs (blue squares) at each offset distance. The mean free path, refractive index, and absorption coefficient of glass were set to 7.84 nm, 1.518, and 0 cm^{-1}, respectively, for the simulation. The mean free path of glass is much larger compared to that of PP, so the occurrence of diffused scattering in the glass wall is much smaller. The trend in these variations is analogous with that of the experimental data; the simulation also finds the same optimal offset distance of 6 mm.

■ CONCLUSION

We have demonstrated the APO scheme, which is able to acquire quantitatively reproducible Raman spectra of housed samples in an oval container as the container orientation is varied with respect to the direction of laser illumination. In contrast, conventional measurements, based on backscattering

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and transmission schemes, are unable to provide acceptably accurate measurements to determine the concentration under equivalent circumstances. We believe that our APO scheme can be widely applied for the nondestructive analysis of housed samples in noncircular (as well as circular) containers by finding the optimal offset distance for each application. We are currently investigating an effective chemometric method to recognize any hidden systematic spectral differences among the APO spectra in order to selectively minimize these to further improve the accuracy.

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Notes

The authors declare no competing financial interest.

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